

Shashank Simha Mysore Ramesh

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EDUCATION

- **Technical University of Munich** Munich, Germany
M.Sc. in Communications and Electronics Engineering; CGPA - 1.7 (1.0 scale) Oct. 2023 – Expected Sept. 2026
- **The National Institute of Engineering** Mysore, India
B.E. in Electronics and Communication; CGPA - 9.08 (10.0 scale) Aug. 2016 – Aug. 2020

EXPERIENCE

- **Infineon Technologies** Munich, Germany
Working Student – Embedded Systems Aug. 2025 – present
 - **Automotive Firmware:** Built motor control and driver software on Aurix MCUs.
 - **Model Based Design:** Developed firmware using MATLAB/Simulink for Aurix hardware.
- **BSH Hausgeräte** Munich, Germany
Working Student – IoT Architecture Oct. 2023 – Jun. 2025
 - **Home Connect Platform:** Developed OEM app using Raspberry Pi for appliance simulation.
 - **Matter Integration:** Integrated Matter spec with ESP32 for smart home devices.
- **Qualcomm** Hyderabad, India
SoC Power Optimization Engineer Jan. 2022 – Sep. 2023
 - **Low-Power Modes:** Built power-saving features for SoCs on Linux/Yocto platforms.
 - **Power Debugging:** Analyzed system power usage with detailed rail breakdowns.
- **PathPartner Technology** Bangalore, India
Embedded Software Engineer Jan. 2021 – Dec. 2021
 - **Android Audio:** Developed audio features for devices using Qualcomm/MediaTek chipsets.
 - **DSP Integration:** Created APIs for on-SoC and external DSP communication.

SKILLS

- **Embedded Systems:** FreeRTOS, ROS2, Linux Device Drivers, Audio/Power Frameworks
- **Model-Based Design:** MATLAB, Simulink
- **Programming:** C, C++, Python, Embedded C, Rust, Android Linux
- **Hardware Design:** VHDL, Verilog HDL, System C, PCB Design
- **Microcontrollers:** ARM and RISC-V microcontrollers and SoCs

INTERNSHIPS

- **Chair of ESI, TUM (Jul. 2025 - Oct. 2025):** Automotive computer vision on AI hardware accelerators
- **Indian Institute of Science (Jan. 2020 - Dec. 2020):** Security architecture in multi-robot environment
- **Indian Institute of Science (Jun. 2019 - Jul. 2019):** RSSI-based trilateration for robot localization

PROJECTS

- **Adaptive Embedded Control System.:** Built Fuzzy-PID system for industrial furnace control with WSN.
- **Smart Speed Governor:** Designed Python-based IoT app on Raspberry Pi to regulate vehicle speed by traffic.
- **FreeRTOS Projects:** Developed real-time features on ESP32 for embedded applications.
- **ECG Data Reduction Algorithms:** Coded AZTEC, TP, DCT to compress ECG data efficiently.

LEADERSHIP EXPERIENCE

- **Mysuru Hub Incharge – IEEE Bangalore Section:** Led and coordinated collaboration events between 5 student branches in Mysore consisting of over 750 students. Won *Outstanding Student Volunteer Award (2019)*
- **Student Coordinator – IEEE India Council:** Supported Webinars and Online Training initiatives. Led a 5-member team to develop the AISYWC'19 website. Won *Outstanding Student Volunteer Award (2020)*
- **Vice Chairperson – NIE IEEE Student Branch:** Led a 13-member team to organize 70+ events, 4 fests, and develop 2 apps. Served as Website Lead for AISYWC'18.

LANGUAGES

- **English:** C1 (*IELTS: 8.0*)
- **German:** A1 (*Actively improving*)

WORKSPACES

- **GitHub:** github.com/shashank-simha
- **Personal Website:** simha.me